

DATA TRANSMISSION

This is the movement of data between the place at which the data is prepared and the computer centre, and between the computer centre and the place at which the results are required.

This means that, you have to link a number of remote terminals to a central computer by use of telecommunication systems.

The basic components of a data transmission system are:

- (a) A central computer.
- (b) Terminal devices.
- (c) Telecommunications link between the central computer & the terminal devices.

Modes/methods Data transmission on a network.

Transmission is possible in 3 modes:

1). Simplex:

This is whereby the transmission of data is possible in only one direction.

2). Half-duplex:

Transmission of data is possible in both directions, but one direction at a time.

3). Full-duplex:

Transmission of data is possible in both directions simultaneously.

Terminal devices:

A *terminal device* is any transmitting or receiving device at the end of a transmission line connected to a computer.

Any peripheral device can be used as a terminal.

Data transmission equipment:

If the terminals are at remote locations, and cannot be connected to the computer by 'landlines', and if postal and messenger services are not suitable, then electronic means of data transmission must be used.

The following are the basic types of Data transmission equipment used to connect the terminals to the computer;

(a) Modem

Where public telephone circuits are used a special equipment (device) called **MODEM** (*Modulator-Demodulator*) must be provided at each end of the line to send data along & receive data from the telephone circuit.

A **Modem** is used to convert (modulate) the bit data into wave data suitable for transmission on public telephone lines. When signal arrives at the receiving end, the waveform must be demodulated back into bit form.

(b) Concentrator (or, Multiplexer)

A Multiplexer is a device used where data from several sources is needed to be send down a single line at the same time.

A *concentrator/multiplexer* is able to use one communication line/link connecting the central computer to a number of terminals that are close to one another, i.e., the data to & from the terminals can be concentrated into a single data stream. This enables the organization to use only one line rather than several, hence reducing line charges.

This process of using one single communications link to carry a number of separate signals is known as **Multiplexing**.

NB: The data is coded in a special way so that it can be sorted out at its destination.

(c) Acoustic coupler:

An **Acoustic coupler** is a computer device that plugs into the computer & the terminal, and has a receptacle/holder into which the telephone handset is placed.

An Acoustic coupler uses the ordinary telephone handset to establish the communication link, and can be used for low-speed transmission (e.g. up to 300 Bits/second).

(d) Dataplex:

A **Dataplex** is used to link a number of terminals which are close together, and then transmit data to the central computer.

Online & Offline data transmission systems:

Any peripheral device (e.g., punched card reader, Magnetic disk drive, Line printer) can be connected via a Modem (or Acoustic coupler) to the central computer.

Online system:

This is where the remote terminal is linked directly to the computer. In this case, data is transmitted via the data link directly into the processor.

Where the online input terminal is slow, the computer must have multi-user capabilities so that while waiting for input, it can link with other terminals and carryout other jobs.

Offline system:

An *Off-line data transmission system* is one where the remote terminal is connected to a storage device such as a Magnetic tape or disk unit, which itself is not linked to the central computer. In this case, the transmitted data is input at a later time.

Offline systems are simpler & less expensive because the terminals are not connected to a computer.

NB: Data transmission controls should be used to ensure that all data sent at one end is received at the other. An example of a control technique, which may be used is the **Echo check**, whereby the data sent is transmitted back to the originating terminal and its accuracy checked.

Carrier Sense Multiple Access: - this is a method of dealing with data collisions on networks, whereby workstations listen for the quiet period on the carrier wave before attempting to send data.

Advantages of using a data transmission communications network.

1. Data collection is faster.
2. Reduces hardware costs. This is because; one large machine can be used instead of a number of smaller ones at different sites.
3. Information kept in the files of the central computer is readily available to whoever may require it and at different locations.
4. Computing facilities are made available to many potential users.
5. Standby facilities are made available in the event of breakdown.

DISTRIBUTED SYSTEMS

This involves using computer systems that have several interconnected processors placed in separate locations.

Each processor has its own local peripherals, e.g., disks, printers, terminals, etc.

Advantages distributing the system.

1. Costs & delays in transmitting and processing data are reduced.
2. There is better local control and service.
3. There is reduced load on the host.

Review Questions

1. (a) What is Data transmission?
(b) State the methods of data transmission on a network.
(c) State the factors to be considered while selecting a data transmission system.
2. Define the following terms:
 - (i). Modem.
 - (ii). Acoustic coupler.
 - (iii). Multiplexer.
3. What is the purpose of a Modem, and where would a modem be used.
4. Explain the term 'Distributed system'.

COMPUTER NETWORKING.

What is computer Networking?

- ❖ *Networking* is the concept of connecting 2 or more computers so that they can share data & resources.
- ❖ It is the interlinking or connecting 2 or more computers for the purpose of communication.

What is a Computer Network?

A *Computer network* is formed when more than one computer is connected together by telephone cables or other media such as Satellites so that they can share information.

Usually there can be from 2 to hundreds or even thousands of computers on the network. Apart from computers, other devices such as Printers, plotters, fax machines, modems, etc can also be connected to the network.

Before then, data was shared by the method of **Sneakernet**, in which data is copied to a floppy disk and carried to another computer.

Note. A computer that is not connected to other computers is a **stand-alone system**.

Advantages of using networks.

1. There is sharing of resources, e.g., computers, data & information.
2. It is able to provide local facilities without the loss of central control.
3. There is even distribution of work, processing load, etc.
4. Improvement in communication facilities.
5. Provides an effective system of backing up data, i.e. in case the hard drive crashes or all files become corrupt, a backup of these files available else where can be used to reconstruct the original files, or else all the work will be lost.

TYPES OF NETWORKS

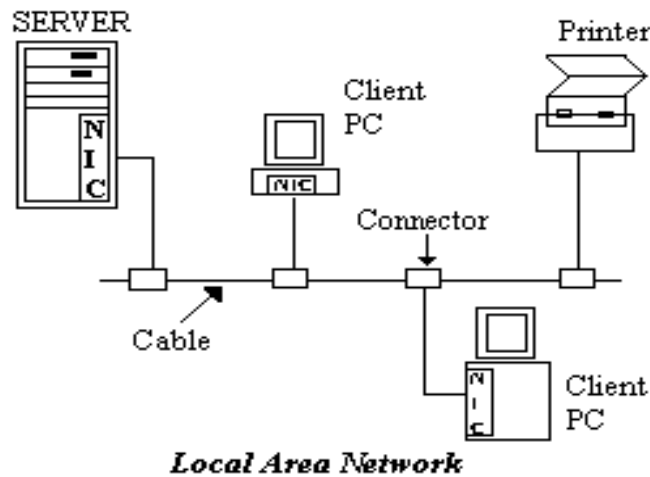
LOCAL AREA NETWORKS (LANs).

This involves interconnecting computers in a single room, rooms within a building or buildings on one side.

Network Components (requirements/features).

The following are the major components of a LAN:

1. Servers – a central computer with high processing speed and storage capacity.
2. Client Workstations/ terminals.
3. Network Interface cards (NICs).
4. Network software/ Network Operating systems.
5. Peripheral devices.
6. Network cables & media.
7. Network Accessories
8. Network Resources, e.g., MODEMS.
9. ISP (Internet Service Provider), if the network is Worldwide.



Server.

Computer networks usually have one computer reserved as the **Server** or “**Mother**” of all computers on the network.

A **Server** is a powerful computer that provides services (shared resources) to the other computers on the network. Usually the Server has a higher hard disk & main memory (RAM) capacity than the other computers on the network.

It is used to store & run the network operating system.

The major function of the Server is to enable information, resources & network devices to be shared by users on a computer network.

Usually, there is only one server in a LAN.

Network Operating System.

The Network operating system enables all resources in a network to be shared.

Typical Network operating systems include;

- ⌋ Novell Netware.
- ⌋ Windows NT.
- ⌋ Unix, and
- ⌋ Linux.
- ⌋ All versions of Windows except Windows 3.1 and 3.11

Clients (workstations)

Workstations are Personal Computers (PCs) attached to the network, on which the network users do their work.

These computers (Clients) are usually less powerful than the server, and use the resources provided by the Server.

The Workstations have their own operating systems and files. The PCs can be IBM or compatible running MS-DOS, OS/2, Windows, etc.

Network Interface Cards (NICs).

NICs enable each computer on the network to be connected to the network cable.

They are usually installed in the expansion slots in the Server and the workstations.

Every card must have network *card driver software* loaded in the computer, which provides the communication between the card and the network operating system software.

The performance of the NIC depends on the *card bus width* & the *speed of the card*.

Network Cables.

These are the media used to physically connect computers and other devices together in a network. They act as the paths on which data signals travel from one computer to the others in the network.

Network Accessories.

These are the various kinds of connectors used for connecting the cables together or to the network interface cards on the computers.

Network Resources.

A network **Resource** is anything available to a client on a network (i.e. what is to be used by network users).

Examples of resources that can be shared:

- ❖ Data.
- ❖ Messages.
- ❖ Graphics.
- ❖ Fax machines.
- ❖ Network Printers.
- ❖ Modems (**M**odulator-**D**emodulator).
- ❖ Files.
- ❖ Disk space.
- ❖ Computer processing power.
- ❖ Communication ports.

Shared data: -These are the files or software provided by Servers across the network.

User: - Any person that uses a client to access resources on the network.

Network Protocols.

- ❖ These are written rules & procedures for communication.
- ❖ They are languages that computer uses to talk to each other over network.
- ❖ They are the standard set of rules describing the transfer of data between devices. It explains the format of the data and the signals to start, control and end transfer.

They originate from the network administrator, e.g. Logging Off, Network Logins, etc

Review Questions

1. Define the following terms:
 - (a) Data communication.
 - (b) Computer Networking.
2. State and explain the THREE types of Networks.
3. (a) Define the term Protocol as used in computer networks.
(b) Give THREE examples of Network software.
4. Explain three advantages and three disadvantages of networking computers.
5. Explain the meaning of the following terms as used in computer networks.
 - (i). Server.
 - (ii). Client.
 - (iii). Resource.
 - (iv). User.
 - (v). Peer.

Transmission Media.

The transmission media is the physical path between the transmitter and the receiver in a data transmission system.

The characteristics and quality of the data transmission are determined by: *the nature of the signal & the nature of the medium.*

The most significant impairments are:

- ⌋ Attenuation distortion (attenuation → decrease/reduction/lessening; distortion → bend/twist).
- ⌋ Delay distortion, and
- ⌋ Noise (thermal, intermodulation, crosstalk, impulse, etc).

There are 3 major types of transmission media used in LAN, namely:

- (i). Twisted pairs.
- (ii). Coaxial cables.
- (iii). Optical fibre cables.

Twisted Pair.

A twisted pair consists of two insulated copper wires twisted in a spiral pattern to minimize electromagnetic interference.

Twisted pair is the cheapest media & the most common transmission medium for both analogue and digital signals. It is the backbone of the telephone system & the workhorse for intrabuilding communications.

Coaxial Cable.

The coaxial cable consists of two conductors. The conductors are insulated and shielded to provide high noise immunity (protection/resistance).

The coaxial cable has been the most versatile (all-round/flexible/widely used) transmission medium in telecommunication systems & Local area networks. However, today its role has been taken over by UTP and Optical fibre cables.

Optical Fibre cables.

Optic fibre cables are made of glass fibres and transmit information in light form.

An optical transmission system consists of the transmission medium, the light source and light detector.

Optical fibres offer high bandwidth, low signal loss & secure data transmission at a reasonable cost.

LAN TOPOLOGIES.

Topology refers to the:

- ❖ Way the end stations or nodes of the network are interconnected.
- ❖ Ways/methods of interconnecting computers.
- ❖ Arrangement of computers, cables & other networking components.
- ❖ Shapes formed by a network.

It is defined by the layout of the communication links & switching elements, and it determines the data paths that may be used between any pair of stations.

There are 4 types of topologies used:

- (a). Star topology.
- (b). Ring topology.
- (c). Bus topology.
- (d). Mesh topology.

TERMS.**(a). Signal Sending Process (SSP).**

Data is sent to all computers in the network but only the destination computer accepts.

Only one computer at a while can send messages.

The number of computers on the network affects the capability of SSP.

Here, each computer must have identification to allow for computer specification.

(b). Signal Bounce (SB).

When a signal is sent to enter the network, it travels uninterrupted from one end of the cable to the other, hence, keeps bouncing back and forth thus preventing other computers from sending signals.

The signal must therefore be stopped after it has had a chance to reach the proper destination.

(c). Terminator.

A component/ device placed at the end of cable to absorb the signals. This clears the cables so that other computers can send data.

(d). Disruption of Network communication.

Communication will be broken if the cable is disconnected. Network activity stops & the network is said to be down (not operational).

LAN PERFORMANCE.

There are 5 factors that affect the performance of a LAN.

1. Processing power of the Server.
2. Hard disk and SCSI Controller.
3. Network Interface card.
4. Cable speed.
5. Number of concurrent users.

Processing Power of the Server.

The processing power of the server determines how fast it can process a network request.

The factors that affect the processing power are:

- ❖ Speed and number of processors.
- ❖ RAM size.

RAM requirement is affected by:

- The number of Users. For example, for every concurrent user on the network, add 1MB.
- The applications being run, and
- The database engine.

Hard disk / SCSI Controller.

The hard disk of the server contains the network operating system and the database. Every time a user requests for data, the hard disk is involved.

The hard disk is connected to external devices through the SCSI (**S**mall **C**omputer **S**ystem **I**nterface) controller card.

The performance of a hard disk can be determined by factors such as:

- (i). Its size.
- (ii). Seek time.
- (iii). Data transfer rate through the SCSI controller card bus/RAM.

The present high performing hard disks are 9GB in size with Seek times of 6.5ms.

The performance of the SCSI controller card depends on the computer speed and bus width. The current high performing SCSI is the Ultra fast wide SCSI-4 with 32-bit bus width and over 100MBps transfer rate.

Network Interface Card.

The bus width of the network interface card affects the speed at which it responds to requests. The wider the bus width, the better the performance. The current highest performing cards are 64-bit, 100MBps cards.

You can also improve the performance by splitting the network load among two or more network interface cards.

Cable speed & Number of concurrent Users.

The bandwidth of the network cable determines the maximum no. of requests that can be serviced concurrently and the speed of data transfer.

The network cable is usually shared by the number of users on a network.

For example, for a 100MBps cable with 100 users, the average speed available to each user is approx. 1MBps (i.e. Total speed/No. of Users).

WIDE AREA NETWORK (WAN)/LONG HAUL NETWORKS (LHN).

WANs are formed whenever computers or networks are connected over a long distance. The computers connected could be in separate cities or countries.

Whenever an individual or company wishes to connect to a distant computer or network, they normally use the existing telephone network. This eliminates the cost of laying new long distant communication lines.

The transmission system in the telephone network could be a *Cable* for short distances, a *VHF* (Very High Frequency) or *UHF* (Ultra High Frequency) *radio* for moderately long distances, *Microwave radio* for very long distances and *Satellite communication systems* for international communication.

However, the telephone network is designed for analogues signals (voice or sound waves) and not data (digital) signals. A data **Modem** (modulators/demodulators) must therefore be used to connect the computer to the telephone network.

The Modem converts the digital signal to analogue form (**modulation**) before transmission and then **demodulates** it at the receiving end (i.e. converts the analogue signal back to digital form).

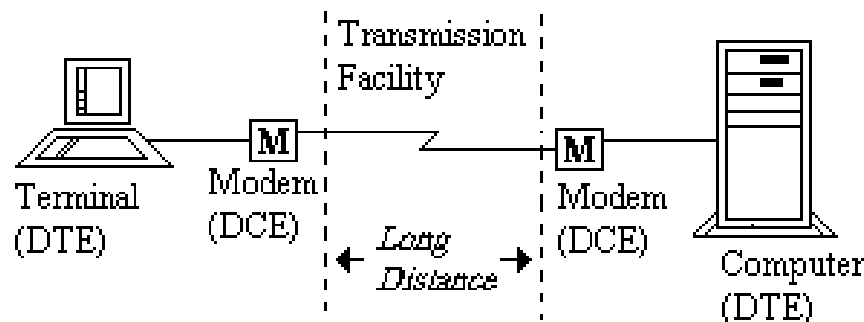
Data Terminal Equipment (DTE) & Data Circuit-terminating Equipment (DCE).

Data Terminal Equipment (DTE) refers to the equipment used for originating or terminating data.

They include;

- ☞ Computers (Servers),
- ☞ Dummy terminals,
- ☞ File servers.
- ☞ Printers.
- ☞ Terminators, etc.

Data Circuit-terminating Equipment (DCE) is the equipment used to interface DTE equipment with networks, such as Modems.



INTERNETWORKING.

Interconnecting of the various hardware & software products from different manufacturers together into a single network requires that the equipment must be able to **Interoperate** (i.e. they must be able to communicate and work) with each other.

Common standards are therefore required to enable these diverse products to interoperate with each other and to be portable across one another.

In response to this, the **International Organization for Standardization (ISO)** in 1977 developed the Open Systems Interconnection (OSI) reference model that defines uniform methods which enable interoperability between heterogeneous (varied/diverse) systems.

OSI Reference Model.

The OSI reference model groups similar computer communication's protocols according to functions into 7 layers, to facilitate communication between application processes located on different computers.

The layering architecture also makes it easy to modify or replace protocols without adverse effects.

7	Application layer
6	Presentation layer
5	Session layer
4	Transport layer
3	Network layer
2	Data link layer
1	Physical layer



The OSI Reference Model

Physical layer: Transmits raw bits over a physical medium. Converts frames to electronic signals and vice versa.

Data Link layer: Provides interface to a communication medium on the physical layer. Changes packets of data into frames and vice versa. May provide error correction and detection.

Network layer: Serves the transport layer by routing packets between adjacent or local network devices.

Transport layer: Provides reliable transparent transfer of data between end points. Ensures that data units are delivered free of errors, in sequence, with no losses or duplication.

Session layer: Negotiates & manages communication sessions between processes on the network.

Presentation layer: Defines data formats to be exchanged & transforms data required by many applications such as encryption & compression.

Application layer: Provides network services such as file sharing, network transparency, distributed processing, file transfer and network management to the users of the OSI environment.

Internetworking devices.

There are 4 major devices used in interconnecting different networks, namely:

- ✧ Repeaters.
- ✧ Bridges.
- ✧ Routers, and
- ✧ Gateways.

Repeaters.

They function at the Physical layer of the OSI model.

They are used to extend the geographical coverage of a LAN by locally connecting two or more LANs into a single LAN. Repeaters re-time, amplify and repeat all electrical signals carried by the network.

Bridges.

They function at the Data Link layer of the OSI model.

They connect similar types of LANs (i.e. Ethernet to Ethernet, Token Ring to Token Ring) over wide area communication links.

Bridges are used when fast operation is required for relatively small & simple networks, and cost is critical.

Routers.

They function at the Network layer of the OSI model.

They maintain routing tables that contain the network address in the data packet supplied to it, to look up internal address tables and the possible routes to the destination. The best path can then be selected on a packet by packet basis, on speed, cost, and delay considerations.

A Router is used to link dissimilar LANs (e.g. Ethernet to Token Ring) for large/complicated networks with heavy traffic and where cost is not put into much consideration.

Note. Routers are special-purpose computers, and have operating systems of their own. Hence, they can be managed and configured for optimum performance.

Gateways.

They function in all the seven layers of the OSI model, since it allows any terminal to connect to any computer on the network.

A gateway does everything that the Router does but for one specific application only. For example, linking one LAN to a proprietary Wide Area Network.

Advantages/ importances of networking computers.

1. It makes the sharing of data much easier and more efficient for users.
2. It makes users more productive as several people can enter, evaluate and process data at the same time.
3. Resources such as Printers, fax devices, modems can be shared by the users.

4. Gives the user the ability to access powerful computers such as Mainframes from a remote area.
5. It facilitates fast and effective communication, e.g., E-mail.
6. Increases data security by restricting physical access and using high quality Servers.

Problems of networking computers.

1. It is expensive to construct and maintain.
2. Viruses spread through networks.
3. Computer crimes are common.
4. Security of data may not be guaranteed.

Review Questions

2. (a). What is a Local Area network?
(b) State the LAN topologies.
3. (a). What does the term Topology refer to as used in computer networks.
(b). With the use of clean diagrams, explain the following network topologies.
 - (i) Bus.
 - (ii) Star.
 - (iii) Ring.
 - (iv) Mesh.
4. (a). What is the primary advantage of star topology?
(b) State THREE advantages and TWO disadvantages of a Mesh topology.
5. List any THREE items that may be referred to as Data Terminal Equipment in a Network.
6. Draw the OSI Reference Model and describe the layers. Explain how bridges and routers work.
7. What is the function of a Gateway, and where is it used?
8. What are some of the problems of computer networking?

USING THE NETWORK.

A **Network** is a group of two or more computers connected to each other by a cable, over telephone wires or through wireless communication, or to a central Server so they can share Resources such as documents and Printers.

By connecting to a Network, you can: -

- ❖ Use Programs and documents from another computer without passing floppy disks back and forth.
- ❖ Print documents on a Printer attached to another computer, or use another Computer's fax modem, just as if they were connected to your computer.
- ❖ Gain access to the Internet.
- ❖ Send and receive messages by using electronic mail (e-mail), or connect to your computer from home.

To install a modem.

1. Click **Start**, point to **Settings**, click **Control Panel**, and then double-click **Modems**.
2. Follow the instructions on your screen.

Note. If the **Modems Properties** dialog box opens instead of the Install New Modem wizard, click **Add** to start the Install New Modem wizard.

CONNECTING TO A NETWORK.

If your computer is set up to use a network, Windows will prompt you for a network password at startup, and the **Network Neighborhood** icon appears on the desktop.

To connect to another computer on your network.

1. On your desktop, double-click **Network Neighborhood**.
2. Double-click the computer you want.
If you don't see the computer you want, double-click **Entire Network**.

Note. If the Network Neighborhood window is empty or the icon is missing, networking is not available.

Setting up your computer to connect to the network.

There are 2 major steps in setting up a network:

- (a) Setting up your hardware, and
- (b) Setting up your software.

Note. Before setting up the software, be sure your network hardware is correctly installed. This includes the Network adapter (network card), cables or any other device.

1. Open the **Control Panel**, and then double-click **Network** to open the **Network** dialog box.
2. Click **Add**.
3. Click **Adapter**, and then click **Add**.
4. Follow the instructions on the screen. If you don't know what kind of adapter you have, check the documentation that came with it.
When you set up a network adapter, Windows automatically sets up the other network components you need to use the network.

Note. If you have a network Client that is not listed (but is written specifically for Windows 98), click **Have Disk**, and then follow the instructions on the screen.

For example, with the Microsoft Client for NetWare Networks that is included in Windows 98, you can connect to the Novell Netware servers.

To install this client,

1. Open the **Network** dialog box, then click **Add**.
2. Click **Client**, and then click **Add**.
3. In the **Manufactures**, click **Microsoft**.
4. In the **Networks Clients**, click **Client for Netware Networks**.

To install the Novell Netware workstation client,

1. Open the **Network** dialog box, then click **Add**.
2. Click **Client**, and then click **Add**.
3. In the **Manufactures**, click **Novell**.
4. In the **Networks Clients**, click the version of the Novell Netware you want.

Client is a computer that accesses shared network resources provided by a central computer (**Server**), on a Local Area Network or the Internet.

A client computer uses files, printers & other resources shared by the server.

To specify your computer and workgroup names.

After setting up network software, you need to give Windows some information to identify your computer on the network. E.g., your network administrator may have assigned your computer to

a workgroup of computers. These are likely to contain most of the resources you will want to use.

To identify your computer on the network;

1. Open the **Control Panel**, and then double-click **Network**
2. Click the **Identification** tab.
3. Type your computer name, your Workgroup name, and a brief description of your system that other people on the network will see when they look at the listing of network computers.
The computer name must be unique. You cannot use a name already in use on the network.
4. Click **OK**.

Workgroup is the group of computers that your computer is in.

A workgroup is composed of the computers you are most likely to communicate with and those which contain most of the network resources (such as files and printers) you will want to use.

NETWORK COMPONENTS.

- (a). **Client Software** enables you to connect and use network resources such as Folders and Printers that have been shared on computers by other people on the network.
- (b). **Service Software** includes services such as File and Printer sharing for your computer or automatic backup to a network Server and enable you to share your resources with other network users.
- (c). **A Network Adapter** is an expansion card or other device that physically connects your computer to the network.
- (d). **A Protocol** is the “language” your computer uses to communicate with other computers on the network.
Several protocols are available. Two computers must use the same protocol to communicate with each other.

Network card is hardware that you insert to a computer to connect the computer to a network.

To log on to the network.

1. Click **Start**, and then click **Log Off**.
2. Click **Yes**.
3. In **Enter Network Password**, type your user name and password.

Note. If you want, you can make your Windows password and your network password the same.

Network password is a password you use to log on to a network. You can also change your settings so you can log on to network without a password.

To Log off the Network.

1. Click **Start**, click **Log Off**, and then click **Yes**.
This closes all your programs, disconnects your computer from the network, and prepares your computer to be used by someone else.
2. To use the computer without connecting to the network, in the **Enter Network Password**, click **Cancel**.

To let another user log on to the network from this computer, in the **Enter Network password**, enter the new user’s name and password when prompted and then click **OK**.

To change your network password.

1. Open the **Passwords Properties** dialog box, then click **Change Other Passwords**.
2. Click the password you want to change, and then click **Change**.

3. Type your old password.
4. Type your new password, and then type it again in **Confirm New Password**.

A **server** is a computer on a Local Area Network that is running software for controlling access to all or part of the network and its resources.

A Server stores and controls shared resources that can be used by other computers on the network called **Clients**.

To change your password for a NetWare server.

5. Connect to your preferred Server.
6. Click **Start**, point to **Programs**, and then click **MS-DOS Prompt**.
7. Change to the drive mapped to your preferred server.
8. Change to the folder that contains the SETPASS utility.
9. Type your old password.
10. Type your new password.

Notes.

-) If you do not know where the SETPASS utility is, use the **Find** command on the **Start** menu to locate it.
-) If your password has already expired, you cannot log on to the network to change the password. Contact your network administrator.

WORKING ON A NETWORK.

To find a computer on your network.

- (i). Click **Start**, point to **Find**, and then click **Computer**.
- (ii). If you know the name of the computer you are looking for, type it in **Named**.
For example: *marketing*
- (iii). If you know the path to the Shared folder you are searching for, you can specify both the computer and folder name.
For example: *\\marketing\reports*
- (iv). Click **Find Now**.

Note. You can also find a computer by double-clicking **Network Neighborhood** on the desktop.

Sharing your Folders or Printers.

If you installed file and printer services for Microsoft Networks or NetWare Networks, you can share your documents and any printers attached to your computer with other people on the network. To share documents, you share the folder they are in.

Shared Resources: - An item on another computer, such as a folder, file, or printer that a person has made available for you to use on the network.

Shared Folder: - A folder on another computer that is available for you to use on the network.

Giving permission to use your shared resources.

There are 2 ways to grant people access to the folders and printers you share over the network.

- ❖ *Share-level access control* – anyone wanting to use the shared resource must know the password you have assigned to it.
- ❖ *User-level access control* –you specify the names of people or groups who are authorized to use a shared resource. If you choose this type of access, you need to specify the computer or domain where the list of authorized users is located.

Windows automatically assigns a permission setting when you set up to share resources.

To see what network Printers and Folders are available.**(i). Double-click Network Neighborhood.**

To see additional printers and folders that are available, double-click **Entire Network**.

(ii). Double-click the computer whose Shared resources you want to view.

Note. If you can't see a resource on the network, you might not have access permission.

To enable file and printer sharing on your computer;

1. In **Control Panel** double-click **Network** to open the **Network** dialog box.
2. Click **File and Print Sharing**.
3. Select the check box for the sharing options you want.
A check mark indicates the feature is activated.
4. Click **OK**.

To share your printer.

1. Click **Start**, point to **Settings**, and then click **Printers**.
2. In the Printers window, click the printer you want to share.
3. On the **File** menu, click **Properties**.
4. Click the **Sharing** tab, and then click **Shared As**.

Notes.

-) If the **Sharing** tab is not visible, you need to enable file and print sharing services.
-) The Tabs that appear in the Printer properties dialog box will vary depending on the type of printer you have.
-) You can only share a printer that is connected to your computer.

Using Resources located on other computers.

Working with a file or folder located on another computer is just like working with those on your own computer.

To use a shared network Printer.

- (i).** Double-click **Network Neighborhood**, and then locate the computer with the printer you want to use.
- (ii).** Double-click the computer with the printer you want, and then click the printer icon.

To use a printer that has been shared on the network, you need to set up the printer on your computer. To set up the printer, follow the instructions on the screen.

Notes.

-) After you have set up a network printer, you can use it as if it were attached to your computer.
-) To see which computers have shared printers, on the **View** menu, click **Details**, and look for printer names or descriptions in the **Comment** column.

Shared (network) printer: - A printer attached to another computer that has been made available for you to use on the network.

CONNECTING TO A NETWORK FROM A REMOTE LOCATION.**Using Dial-Up Networking.**

Dial-up Networking: -A feature that allows you to use a phone line & modem to connect to another computer.

If you have a modem, you can connect to another computer or to your corporate network by using your phone line.

With Dial-Up Networking, you can have access to information on another computer, even if your computer is not on a network.

The computer you are dialing in to must be set up as a network server for you to use its shared resources. Both your home computer and the network server must have modems installed.

To connect to another computer or corporate network, you need to install your modem and then configure the connection.

To set up a network connection by using Dial-Up Networking;

- (i). Double-click **My Computer**, and then double-click **Dial-Up Networking** to start the New Connection wizard.
- (ii). Follow the instructions on the screen.

Notes.

-) You can also start Dial-Up Networking by clicking **Start**, point to **Programs**, point to **Accessories**, point to **Communications**, and then click **Dial-Up Networking**.
-) To dial a connection that you have already set up, click its icon in the Dial-Up Networking window.
-) To connect to an online service such as a bulletin board, use *HyperTerminal* rather than Dial-Up Networking.
-) If you have already set up a Dial-Up connection, you can click **Make New Connection** to create a new connection.
-) After you connect to another computer, you can see files and folders on the remote computer only if the files and folders are shared.
-) If the Dial-Up Networking folder is not in **My Computer** or the **Accessories** menu, it is not installed.

To install Dial-up Networking;

1. Open the **Control Panel**, and then double-click **Add/Remove Programs**.
2. Click the **Windows Setup** tab, click **Communications**, and then click **Details**.
3. Click the **Dial-up Networking** check box, and then click **OK**.
4. Click **OK** again, and follow the instructions on your screen.

To set up a Dial-Up Networking server.

- (i). Double-click **My Computer**, then double-click **Dial-Up Networking** to start Dial-Up Networking.
- (ii). Click **Connections** and then click **Dial-Up Server**.
- (iii). Click **Allow caller access** to enable others to connect to your computer.
- (iv). Click **Server Type** and then choose a server type from the **Type of Dial-Up Server** list.

Note.

-) Dial-Up Networking and Dial-Up Server must be installed on your computer before you can set up a Dial-Up Networking server.

To change dialing settings for all Dial-Up Networking connections.

- (i). Double-click **My Computer**, and then double-click **Dial-Up Networking**.
- (ii). Click the icon for the connection that has settings you want to change.
- (iii). On the **Connections** menu, click **Settings**.

To change the properties for a Dial-Up Networking connection.

- (iv). In the Dial-Up Networking folder, click the icon for the connection that has properties you want to change.
- (v). On the **File** menu, click **Properties**.

(vi). Click **Configure**.

(vii). Make changes to the modem properties.

Your changes will take effect when you connect to a shared resource that uses this connection, or the next time you open this connection.

Notes.

- ⌋ To change the type of server that you are connecting to, or to specify which protocols the connection uses, click the **Server Types** tab.
- ⌋ You can change settings only if you have already set up a connection.

A Remote User: -A person who connects to a network by using a modem & dial-up networking.

Using Direct Cable Connection to connect to another computer.

With Direct Cable Connection, you can gain access to shared folders on another computer, even when your computer is not on a network. For example, if you have a portable computer, you can use a cable to connect it to your work computer and network.

To set up a direct cable connection;

- (i). Click **Start**, point to **Programs**, point to **Accessories**, point to **Communications**, and then click **Direct Cable Connection**.
- (ii). Follow the instructions on the screen.

Notes.

- ⌋ Once you successfully run Direct Cable Connection on this computer, the Direct Cable Connection wizard does not appear unless you click **Change** in the **Direct Cable Connection** dialog box. The Direct Cable Connection wizard then starts and you can change your previous settings.
- ⌋ If you do not see Direct Cable Connection on the **Communications** menu, it is not installed.

To change the settings for a direct cable connection.

- (i). Click **Start**, point to **Programs**, point to **Accessories**, point to **Communications**, and then click **Direct Cable Connection**.
- (ii). Click **Change**.
- (iii). Make any necessary changes.
Your changes take effect when you connect to a shared resource that uses this connection.

To re-establish an existing direct cable connection.

- (i). On the host computer, click **Start**, point to **Programs**, point to **Accessories**, point to **Communications**, and then click **Direct Cable Connection**.
- (ii). Click **Listen**.
- (iii). Move to the guest computer and repeat step 1.
- (iv). Click **Connect**.

Note. If you have trouble connecting, check the cable.

To view information about your network connections.

Right-click **Network Neighborhood**, and then click **Who Am I**.

To detach from a server by using Who Am I.

- (i). Right-click **Network Neighborhood**, and then click **Who Am I**.
- (ii). Click the name of the server you want to detach from, and then click **Detach**.

Using HyperTerminal

You can use HyperTerminal and a modem to connect to a remote computer, even if the remote computer isn't running Windows. You can also use HyperTerminal to send and receive files, or to connect to computer bulletin boards and other information programs.

To start HyperTerminal;

1. Click **Start**, point to **Programs**, point to **Accessories**, point to **Communications**, click **HyperTerminal**, and then click **Hypertrm.exe**.

Notes

-) For information about how to use HyperTerminal, click the **Help** menu in HyperTerminal.
-) To gain access to files and printers on another computer running Windows, use Dial-Up Networking rather than HyperTerminal.

Using the ISDN Configuration wizard

You can use the ISDN Configuration wizard to set up ISDN to upgrade your analog telephone network to a digital system. A digital system is much faster and more reliable than an analog system.

Note

-) You must have ISDN hardware installed before you can run the ISDN Configuration wizard.

ISDN (Integrated Services Digital Network): - is a high-speed digital telephone service that can dramatically increase the speed at which you connect to the Internet or to your corporate Local Area Network (LAN).

ISDN can operate at speeds up to 128 kilobits per second, which is five or more times faster than many analog modems.

Using Phone Dialer to dial from your computer.

Using Phone Dialer, you can place telephone calls from your computer by using your modem or another Windows telephony device.

To start Phone Dialer;

1. Click **Start**, point to **Programs**, point to **Accessories**, point to **Communications**, and then click **Phone Dialer**.

Note. For information about how to use Phone Dialer, click the **Help** menu in Phone Dialer.

Using NetWatcher to monitor shared resource use

Using Net Watcher, you can see who is currently using resources on your computer. You can also add shared folders and disconnect users from your computer or from specific files.

Before you can run NetWatcher, you must have Client for Microsoft Networks installed, and you must enable file and print sharing options.

To start Net Watcher;

1. Click **Start**, point to **Programs**, point to **Accessories**, point to **System Tools**, and then click **Net Watcher**.

Notes

-) If you do not see Net Watcher on the **System Tools** menu, it is not installed.
-) For information about how to use Net Watcher, click the **Help** menu in Net Watcher.